SOKO SAFI AI

Field Pilot Validation Study & Technical Appendix

Location: Ahero Market, Kisumu County, Kenya **Date:** May – June 2026 **Lead Investigator:** Clare Akoth, Founder

This technical appendix compiles the empirical baseline metrics, technical parameters, and user-validation feedback gathered during a 7-day field pilot deployment conducted at Ahero Market. The objective of this pilot framework was to test the market viability, mechanical storage thresholds, and smallholder financial elasticity for affordable decentralized cold chain solutions before large-scale infrastructure fabrication.

1. Technical Infrastructure & Parameter Customization

To closely simulate the thermodynamic conditions of our future automated solar-powered biodigester hubs, the field team acquired and retrofitted a commercial-grade vapor-compression cooling system.

Thermodynamic Temperature Targeting: The infrastructure's ambient storage environment was tightly regulated and locked to a range of ***10°C – 12°C***.

This specific engineering configuration was chosen based on agronomical research data to delay the physiological respiration cycle of Solanum lycopersicum (tomatoes). By maintaining the temperature strictly above ***10°C***, the system successfully averted cold injury and cellular structure collapse (chilling damage), while concurrently suppressing metabolic decay, transpiration rates, and bacterial soft rot propagation.

2. Field Pilot Cohort & Operational Traction

The pilot study targeted small-scale perishable product handlers, specifically focusing on micro-vendors facing high post-harvest loss vulnerabilities due to local outdoor heat elements. Three distinct traders were tracked closely over a continuous 7-day storage loop:

7 Days	3	KES 20	100%
Continuous Testing Period	Active Pilot Micro-Traders	Tested Price Point Per Crate / Day	Produce Preservation Rate

Participant Identity	Core Market Commodity	Tested Pricing Tier	Retention & Outcomes
Mama Dex	Fresh Tomatoes (Retail)	KES 20 / Crate / Day	100% Retention; successfully eliminated daily inventory dumping loops.
Linet	Fresh Tomatoes & Leafy Greens	KES 20 / Crate / Day	100% Retention; validated severe structural pull for a permanent hub facility.
Baba Elvice	Perishables (Wholesale supply)	KES 20 / Crate / Day	100% Retention; logged intent to scale local inventory volumes safely.

Figure 1: Authentic inventory tracking image from the 7-day Soko Safi field pilot in Ahero Market. Fresh tomato yields remain structurally rigid and completely unblemished inside the customized 10°C - 12°C trial unit, eliminating previous dumping patterns.

3. Qualitative Validation & Field Testimonials

The field survey gathered direct qualitative feedback from the primary market ecosystem actors. Their verified, unfiltered responses highlight the economic relief and viability of the pay-as-you-store pricing strategy:

> *"This is a very good idea... I cannot afford to buy a freezer but I can afford to pay the 20 shillings to store my tomatoes so that I can sell them the following day... In other days I would have thrown them to the pit or given them to chickens because they would rot."*
>
> **— Mama Dex, Retail Trader, Ahero Market**
>
> **INSIGHT: ASSET DEMOCRATIZATION & LOSS ERADICATION**

> *"...When is this hub coming? This is an eye-opener and we will support it fully."*
>
> **— Linet, Perishables Vendor, Ahero Market**
>
> **INSIGHT: GRASSROOTS DEMAND VERIFICATION**

> *"This storage should come... It will enable me to save enough to open my supply system because I'll be sure that the goods are safe and their life extended."*
>
> **— Baba Elvice, Wholesale Vendor, Ahero Market**
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> **INSIGHT: WEALTH ACCUMULATION & VALUE CHAIN EXPANSION**

4. Gender Impact & Community Metrics

Ecosystem baselines indicate that open-air spaces like Ahero Market generate roughly **1,000 kg of organic agricultural waste every day**. More importantly, over 80% of retail food handlers operating in these contexts are women.

By designing a technical database infrastructure (implemented through our automated WhatsApp chatbot flow) that tracks gender metrics natively during the initial user onboarding sequence, Soko Safi AI establishes a continuous **gender-disaggregated data layer**.

As verified by the pilot composition (66% female inclusion), protecting the inventory margins of women immediately increases direct household financial liquidity. This economic security ensures structural support at home, decreases community vulnerabilities, eliminates waste, and directly stabilizes child welfare across the Kisumu County commercial belt.